

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2024

Bachelor in Computer Applications

Course Title: **Advanced Java Programming**

Code No: **CACS354** Semester: **VI**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What is the use of WindowEvent? List different methods of WindowListener interface with its uses.
3. How can you display images in GUI using Java? Explain.
4. Why do we need to create JavaBeans? Explain the steps involved in the creation of a JavaBean.
5. What is the use of PreparedStatement? Differentiate between Statement and PreparedStatement.
6. What are declaration and expression tags in JSP? Explain with examples.
7. What is RMI? Write a RMI client and server program to find the greater number among two numbers.
8. Write short notes on (Any two): a) JMenuBar b) Session handling in Servlet c) Marshalling and Unmarshalling of arguments in RMI

Group C

Attempt any TWO questions[2x10=20]

9. Why do we need to use layout managers in GUI programming? Write a program to create a GUI form that takes data about a customer including cid, name, address, email and age and insert these data into the Customer table of the Shop database.
10. Explain the lifecycle of a Java Servlet. Write a servlet program that takes three numbers from an HTML page and computes the sum of natural numbers up to the greatest number among the three numbers.
11. What is directive element in JSP? Explain in detail. Create a JSP page that verifies the username and password entered from an HTML file and if the username and password are correct it displays a greetings message. Otherwise, it redirects to the register page.